Ground Effect on NACA 2412 Airfoil

Updated Surface-Panel Method, Verification, and Comparison with Legacy Results

March 3, 2026

Abstract

This report studies ground effect on a NACA 2412 airfoil using an updated **surface-panel method** (upper and lower surfaces are both modeled), with image panels for the ground wall. I present: (1) C_L vs. α for multiple h/c, (2) C_p vs. x/c in the requested grouping (one subplot per h/c, all requested α in each panel), (3) γ distribution vs. x/c, and (4) verification against XFOIL for no-ground cases.

I also keep the older camber-line figures in this report for side-by-side comparison.

1 Objective

The goal is to quantify how airfoil height from the ground changes:

- lift coefficient C_L,
- pressure coefficient C_p,
- and circulation/sheet-strength trend (γ proxy).

2 Procedure and Methodology (Updated)

2.1 Airfoil and Cases

- Airfoil: NACA 2412.
- Chord: $c = 1$.
- Height cases: $h/c \to \infty$ (no wall), 1.00, 0.50, 0.15.
- Angles:
  - For C_L: $\alpha = -8^\circ$ to 12°,
  - For C_p and γ: $\alpha = -4^\circ, 0^\circ, 4^\circ, 8^\circ, 12^\circ$.

2.2 Surface-Panel Model

The current solver is a Hess–Smith style 2D surface-panel method:

1. Generate full NACA 2412 geometry (closed trailing edge): upper and lower coordinates.
2. Split the full boundary into panels (cosine spacing in construction, then boundary panels).
3. Unknowns are panel source strengths σ_j and one global vortex strength γ.
4. Enforce no-penetration at every panel control point.
5. Enforce Kutta condition at trailing edge using upper/lower trailing-edge offsets.
6. Solve one linear system for each angle of attack.

2.3 Panel-Method Formulation Used in the New Surface Code

For each case (h/c and α), the solver uses:

- N constant-strength source panels (σ_j, $j = 1, \ldots, N$),
- one global vortex strength γ.

At each control point i, no-penetration is enforced:

$$\sum_{j=1}^{N} A_{ij}^{(\sigma)} \sigma_j + A_i^{(\gamma)} \gamma = -\mathbf{V}_\infty \cdot \mathbf{n}_i,$$

where $A_{ij}^{(\sigma)}$ and $A_i^{(\gamma)}$ are normal-velocity influence coefficients from source and vortex panels.

The trailing-edge Kutta condition is written as equality of tangential behavior between upper and lower TE sides, giving one extra equation:

$$\sum_{j=1}^{N} K_j^{(\sigma)} \sigma_j + K^{(\gamma)} \gamma = -\mathbf{V}_\infty \cdot (\mathbf{t}_{TE,u} + \mathbf{t}_{TE,l}).$$

So each solve is an $(N+1) \times (N+1)$ linear system:

$$\begin{bmatrix} \mathbf{A}^{(\sigma)} & \mathbf{A}^{(\gamma)} \\ \mathbf{K}^{(\sigma)T} & K^{(\gamma)} \end{bmatrix} \begin{bmatrix} \boldsymbol{\sigma} \\ \gamma \end{bmatrix} = \begin{bmatrix} \mathbf{b}_n \\ b_K \end{bmatrix}.$$

After solving for $(\boldsymbol{\sigma}, \gamma)$:

- velocity at near-surface sample points is reconstructed,
- $C_p = 1 - (V/U_\infty)^2$ is computed on upper/lower surfaces,
- total circulation is $\Gamma_{tot} = \gamma \sum_j \Delta s_j$ and $C_L = 2\Gamma_{tot}/(U_\infty c)$.

2.4 Ground-Effect Modeling

For finite h/c, image panels are added across $y = 0$:

- source image has same sign,
- vortex image has opposite sign.

This enforces slip-wall behavior at the ground.

2.5 Post-Processing

- Lift coefficient:

$$C_L = \frac{2\Gamma_{\text{total}}}{U_\infty c}, \qquad \Gamma_{\text{total}} = \gamma \sum_j \Delta s_j.$$

- Surface pressure coefficient from local velocity:

$$C_p = 1 - \left(\frac{V}{U_\infty}\right)^2.$$

- The reported γ plot is a useful circulation-trend diagnostic from upper/lower speed split.

2.6 Verification Workflow

No-ground model outputs are compared with XFOIL (Drela, 1989):

- angles used: $\alpha = 0°, 4°, 8°$,
- compare upper-surface $C_p(x/c)$,
- compute RMSE on overlap of the two x/c grids,
- also compare C_L values.

3 Current Results (Surface-Panel)

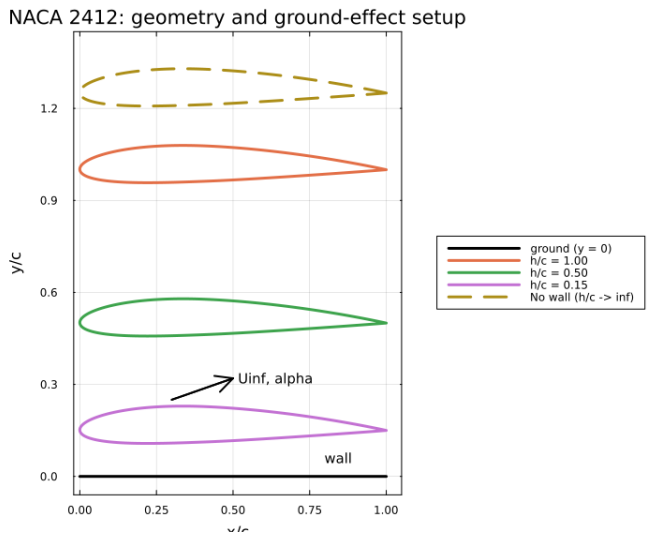

Figure 1: NACA 2412: geometry and ground-effect setup used in the current surface-panel solver.

3.1 C_L vs α

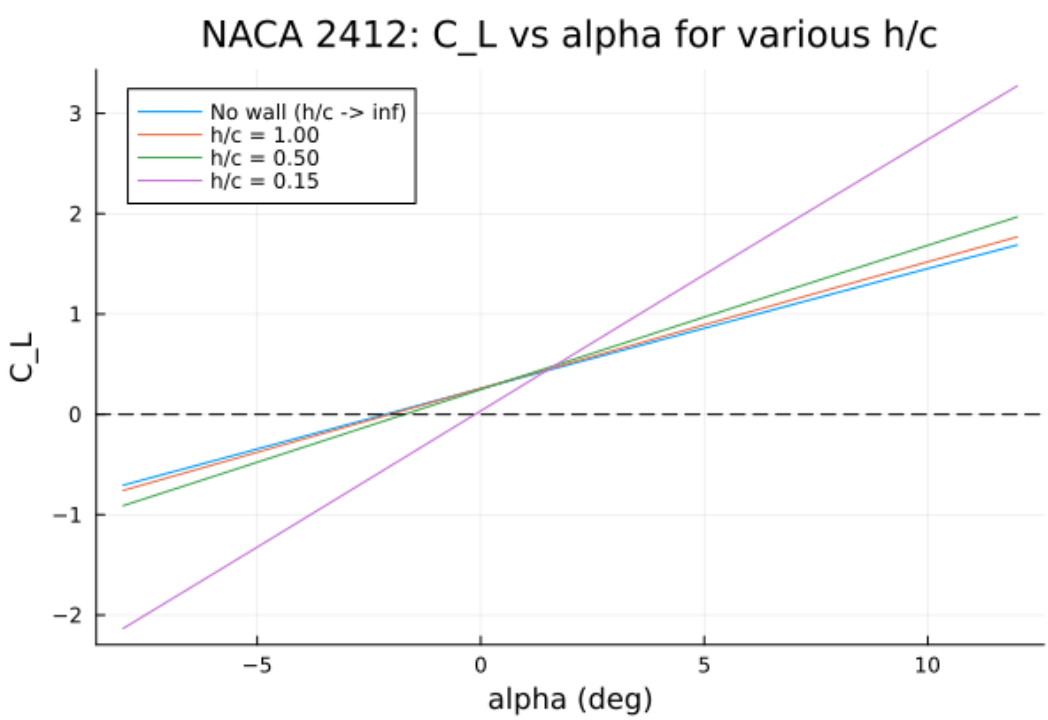

Figure 2: Current result: C_L vs. α for different h/c.

Table 1: Current linear fit from computed C_L curves.

Case	$dC_L/d\alpha$ (per deg)	fitted C_L at $\alpha = 0°$
No wall ($h/c \to \infty$)	0.11993	0.25588
$h/c = 1.00$	0.12658	0.25704
$h/c = 0.50$	0.14414	0.24559
$h/c = 0.15$	0.27086	0.03374

3.2 C_p vs x/c (requested grouping)

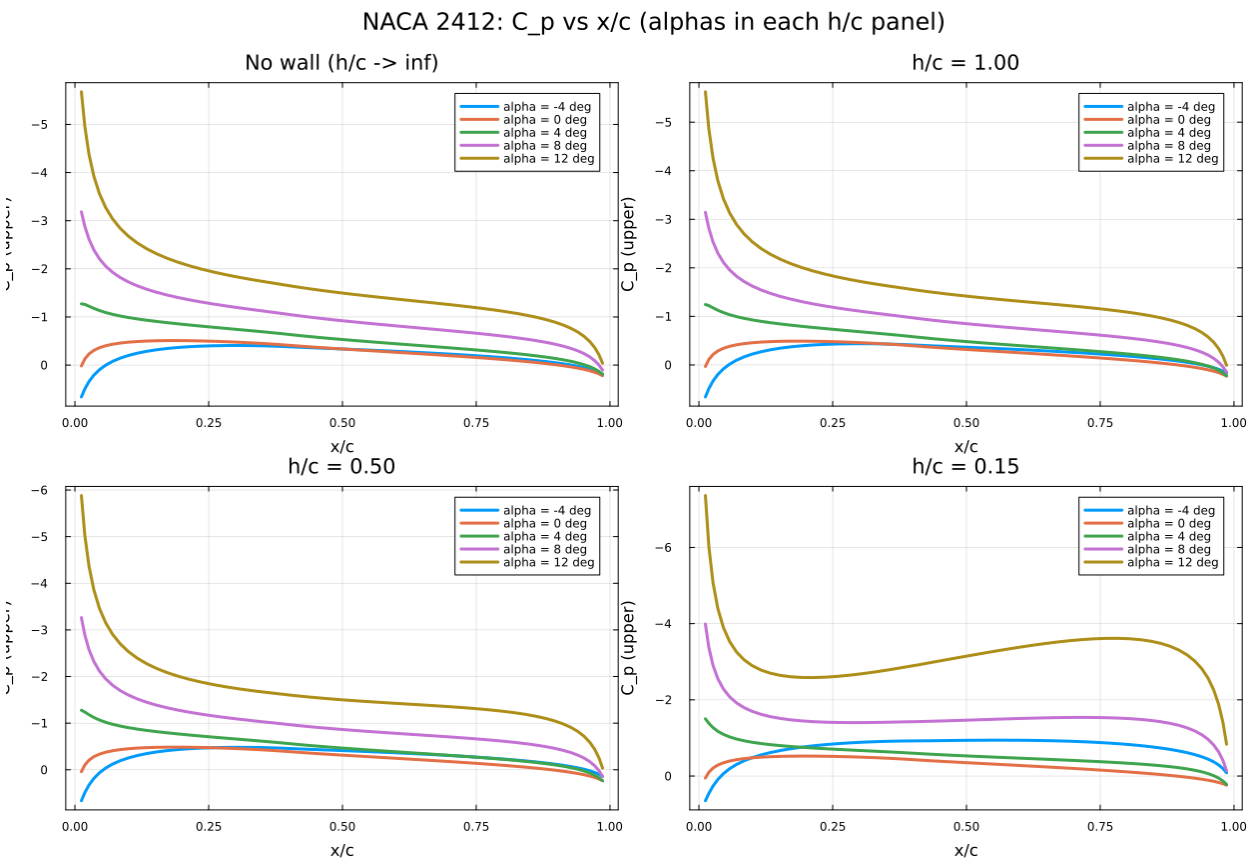

Figure 3: Current result: C_p vs. x/c, one panel per h/c, all requested α curves in each panel.

3.3 γ vs x/c

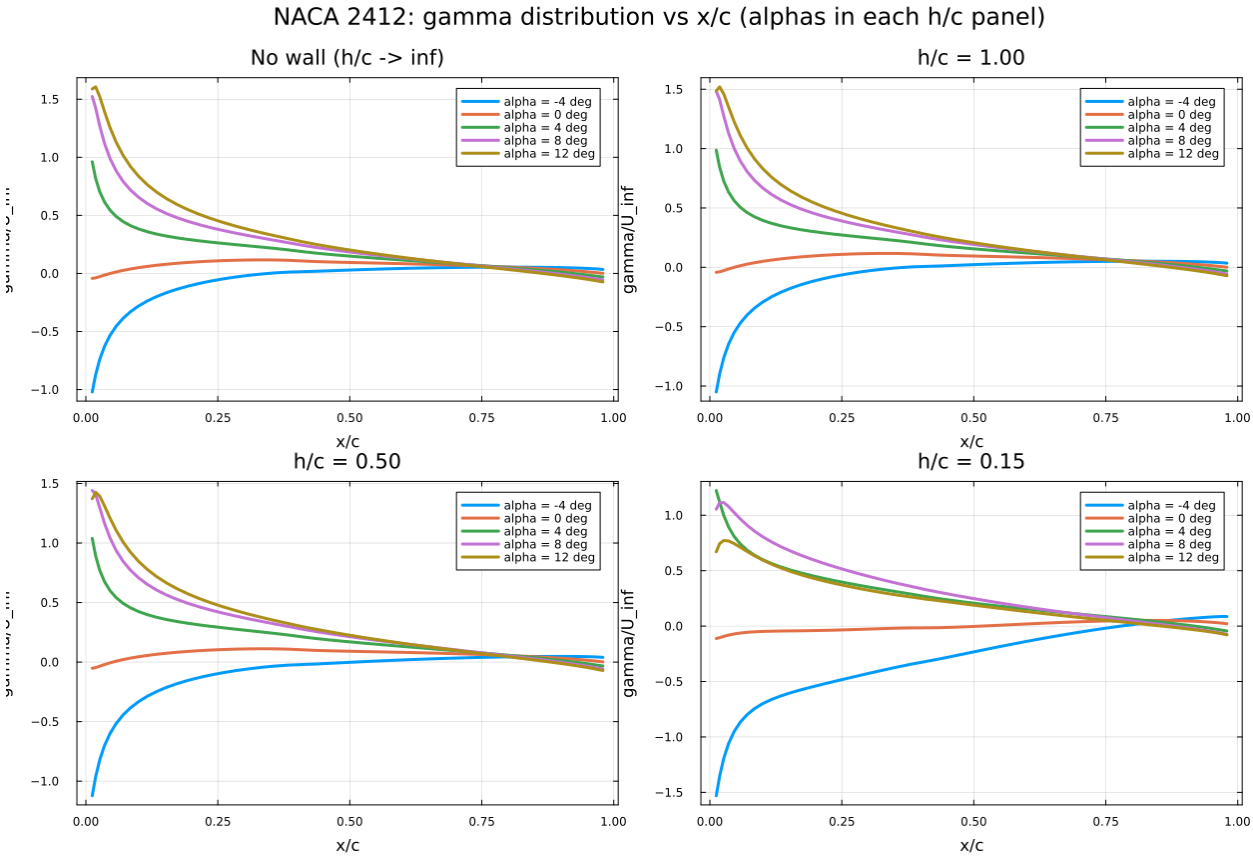

Figure 4: Current result: γ distribution vs. x/c, grouped by h/c.

4 Verification Against XFOIL (Current Surface-Panel Equations)

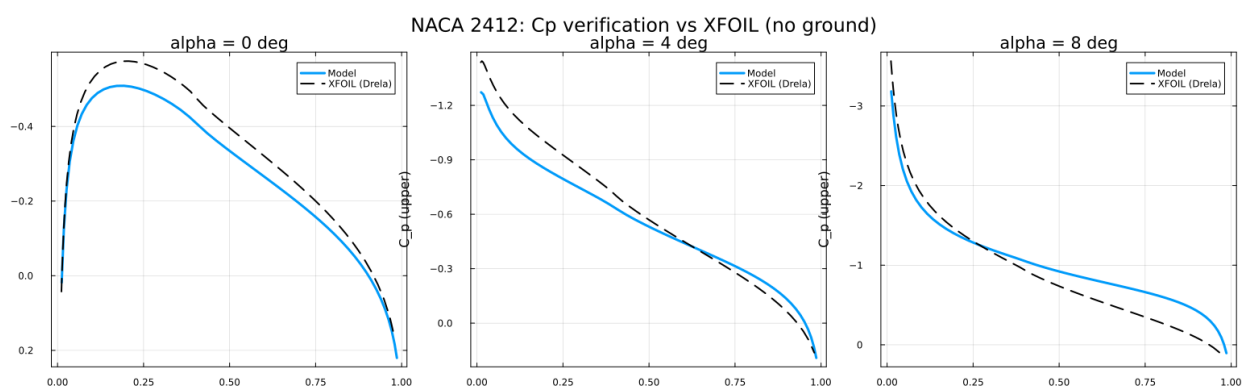

Figure 5: Current verification: model C_p vs. XFOIL C_p (no-ground case).

Table 2: Current per-angle verification metrics (no-ground case).

α	C_p RMSE	$C_{L,\text{model}}$	$C_{L,\text{XFOIL}}$	ΔC_L
0°	0.0489	0.2576	0.2591	-0.0015
4°	0.1077	0.7388	0.7409	-0.0021
8°	0.2366	1.2164	1.2192	-0.0027

Why there is still a small difference now

The difference is much smaller than before, but not exactly zero. This is expected, even with a surface-panel model:

1. **Different numerical formulations:** this code uses one simple low-order panel setup (constant source strength on each surface panel plus one global circulation variable), while XFOIL uses a different panel and wake formulation with its own numerical details.

2. **Trailing-edge and wake treatment are not identical:** small changes in Kutta enforcement and wake modeling can shift both local C_p and integrated C_L.

3. **Leading-edge sensitivity:** C_p gradients are very steep near the leading edge, so small discretization differences create visible local error.

4. **Grid/interpolation mismatch:** RMSE is evaluated after interpolating two different x/c point sets and clipping near LE/TE, which adds small numerical differences.

So the current model and XFOIL are now close, but they are still not mathematically identical solvers.

Error-location and panel-count insights (current)

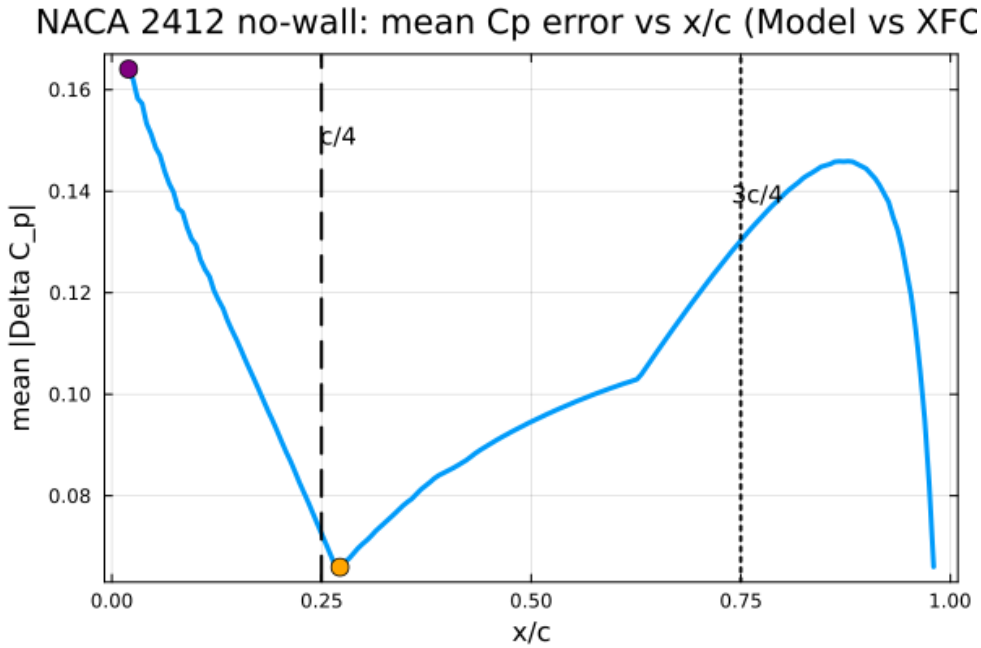

Figure 6: Current mean absolute C_p difference along chord (no-ground, averaged over $\alpha = 0°, 4°, 8°$).

Current statistics from the generated metrics file:

- Mean max error location: $x/c \approx 0.020$,
- Mean min error location: $x/c \approx 0.272$,
- $|\Delta C_p|(c/4) = 0.0726$,
- $|\Delta C_p|(3c/4) = 0.1303$.

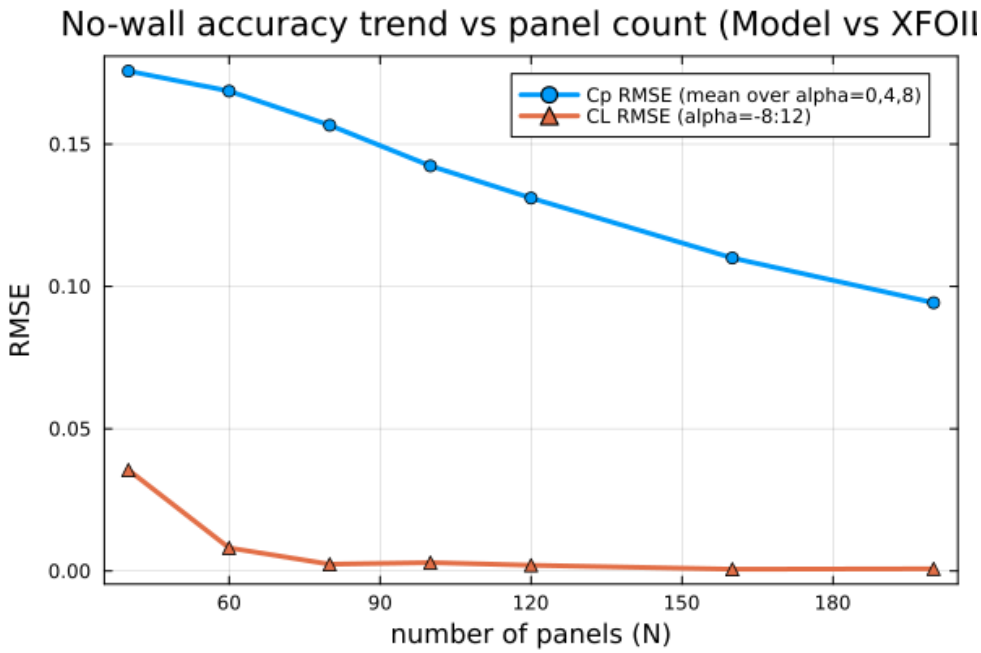

Figure 7: Current no-wall accuracy trend vs panel count (model vs XFOIL).

Table 3: Current panel-count trend (no-ground): C_p RMSE and C_L RMSE.

N	C_p RMSE	C_L RMSE ($\alpha = -8^\circ : 12^\circ$)
40	0.1757	0.0355
60	0.1686	0.0081
80	0.1566	0.0024
100	0.1423	0.0030
120	0.1311	0.0020
160	0.1100	0.0006
200	0.0943	0.0008

Old vs New improvement (camber-line to surface-panel)

To make the improvement explicit, I compared the legacy camber-line metrics with the current surface-panel metrics for the same no-ground verification cases.

Table 4: No-ground verification improvement from old to new model.

Metric	Old (camber-line)	New (surface-panel)	Absolute drop	Relative drop
C_p RMSE at $\alpha = 0^\circ$	0.2719	0.0489	0.2230	82.0%
C_p RMSE at $\alpha = 4^\circ$	0.3326	0.1077	0.2249	67.6%
C_p RMSE at $\alpha = 8^\circ$	0.4117	0.2366	0.1751	42.5%
$\lvert\Delta C_L\rvert$ at $\alpha = 0^\circ$	0.0314	0.0015	0.0299	95.2%
$\lvert\Delta C_L\rvert$ at $\alpha = 4^\circ$	0.0754	0.0021	0.0733	97.2%
$\lvert\Delta C_L\rvert$ at $\alpha = 8^\circ$	0.1191	0.0027	0.1164	97.7%

Table 5: Panel-trend comparison at $N = 120$ (no-ground).

Metric at $N = 120$	Old	New	Absolute drop	Relative drop
C_p RMSE (mean over $\alpha = 0°, 4°, 8°$)	0.3387	0.1311	0.2076	61.3%
C_L RMSE ($\alpha = -8° : 12°$)	0.0851	0.0020	0.0831	97.6%

This confirms the main point: switching to the full surface-panel equations removed most of the large mismatch that came from the camber-line simplification, while a smaller residual difference remains for the numerical reasons discussed above.

5 Legacy Camber-Line Figures (Kept for Comparison)

The following figures are the older camber-line based outputs. They are intentionally kept here so the evolution of the model can be shown in one report.

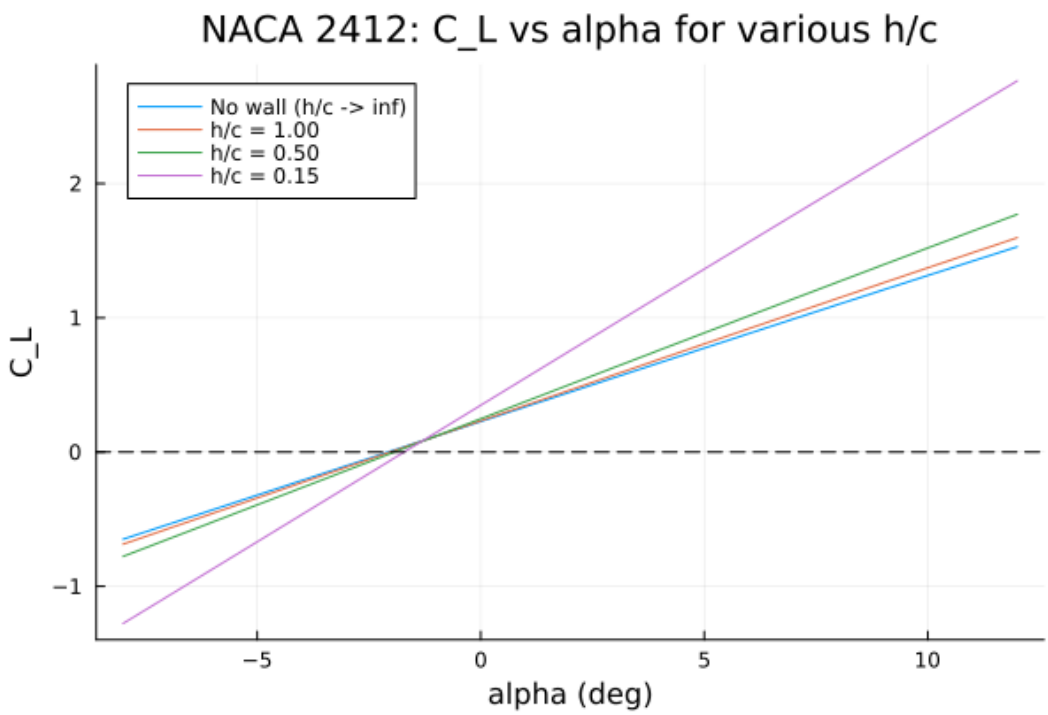

Figure 8: Legacy result: C_L vs. α (camber-line model).

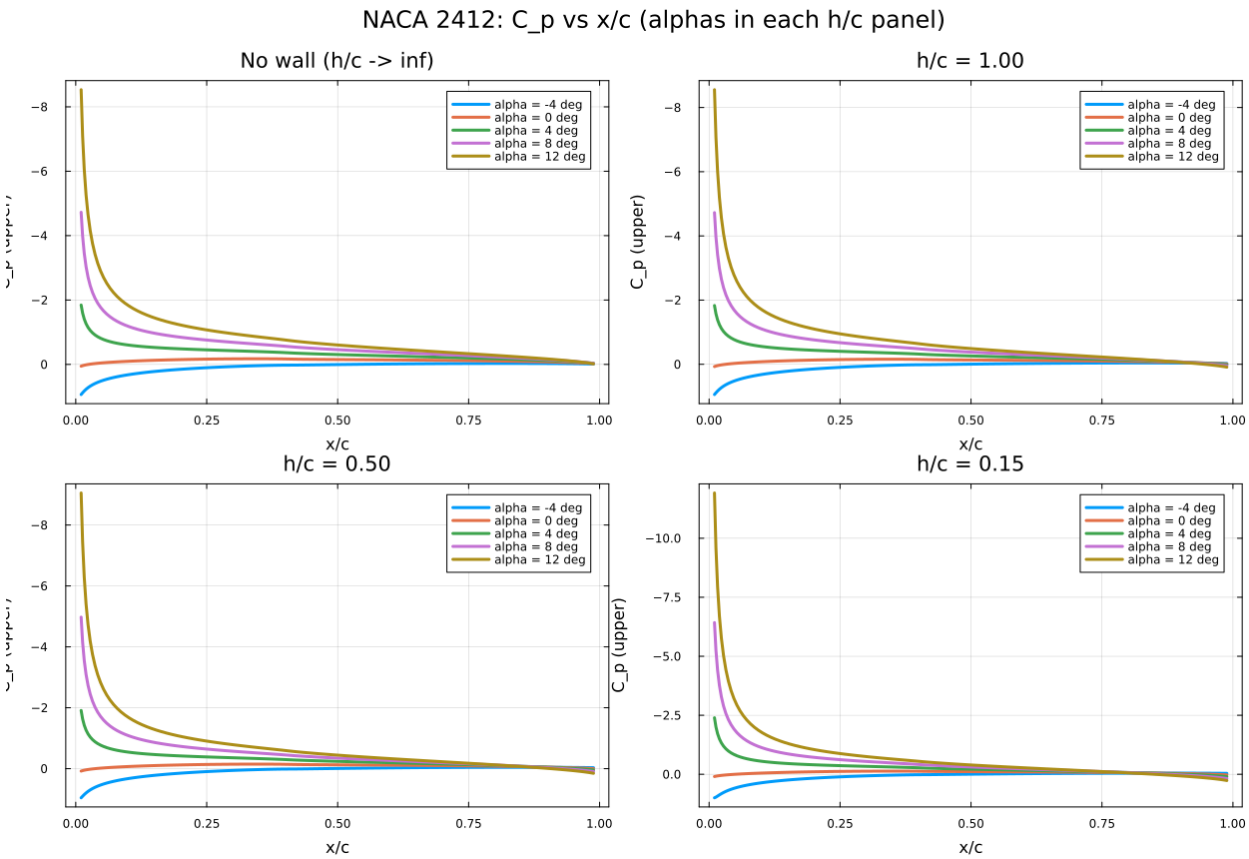

Figure 9: Legacy result: C_p vs. x/c (camber-line model).

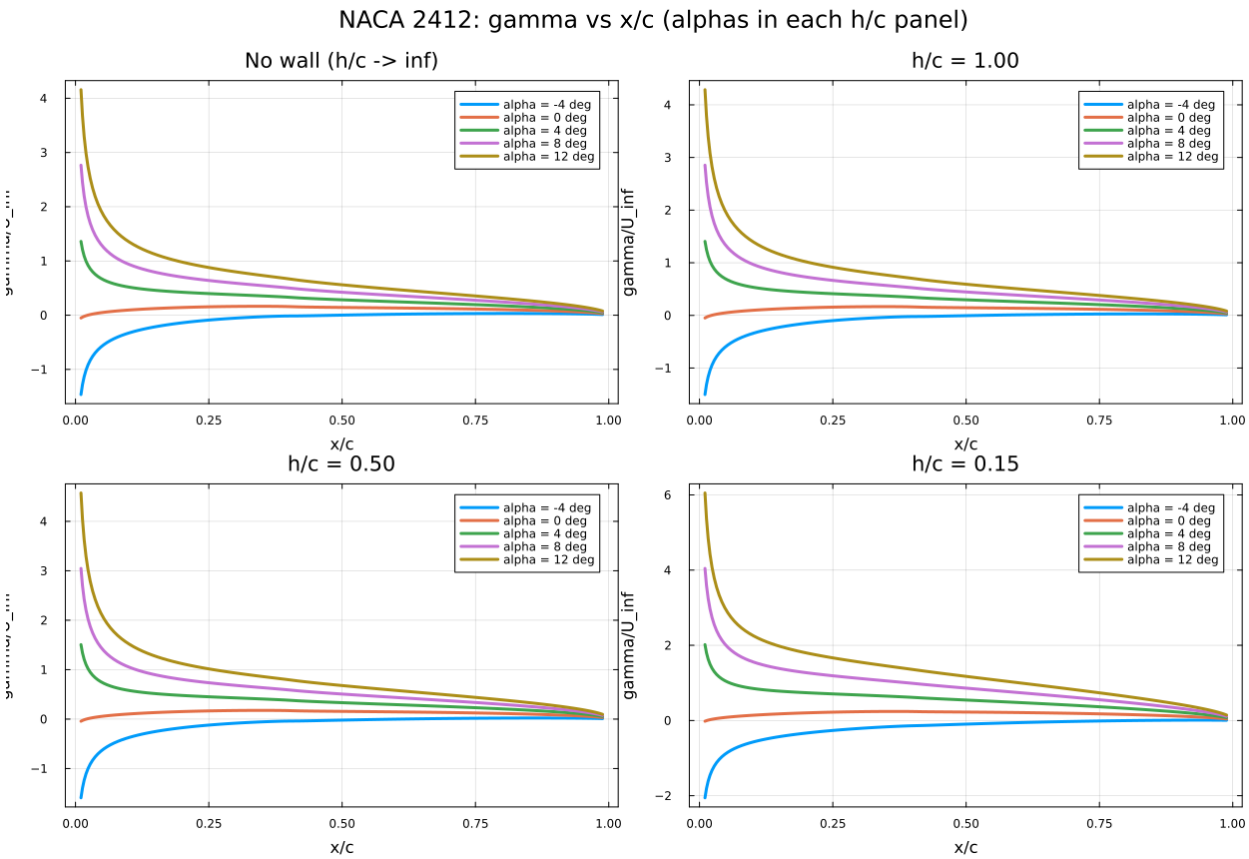

Figure 10: Legacy result: γ vs. x/c (camber-line model).

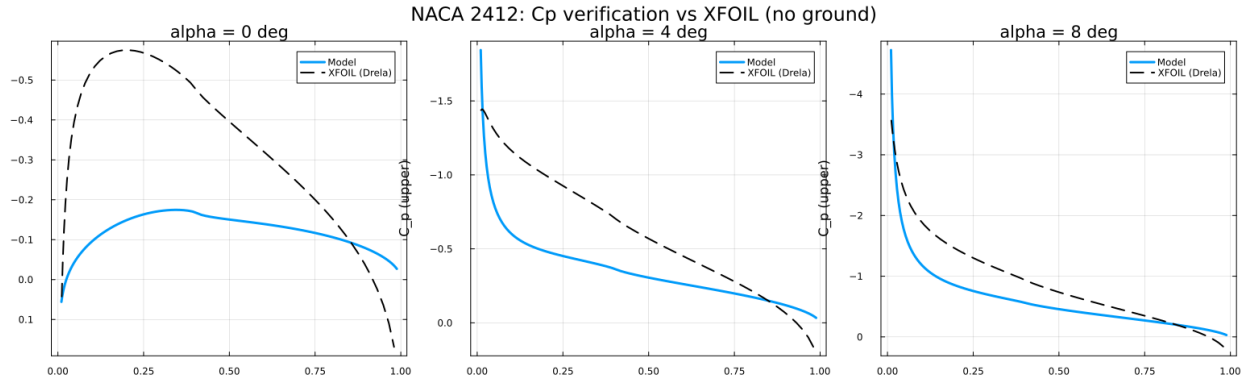

Figure 11: Legacy verification: model C_p vs XFOIL (camber-line model).

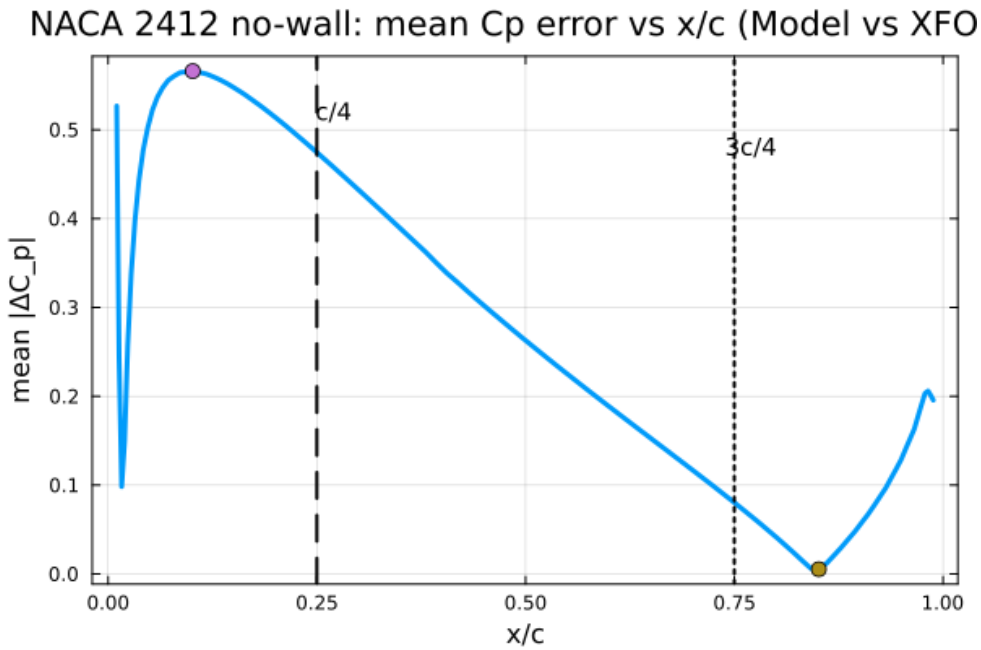

Figure 12: Legacy mean $|\Delta C_p|$ profile (camber-line model).

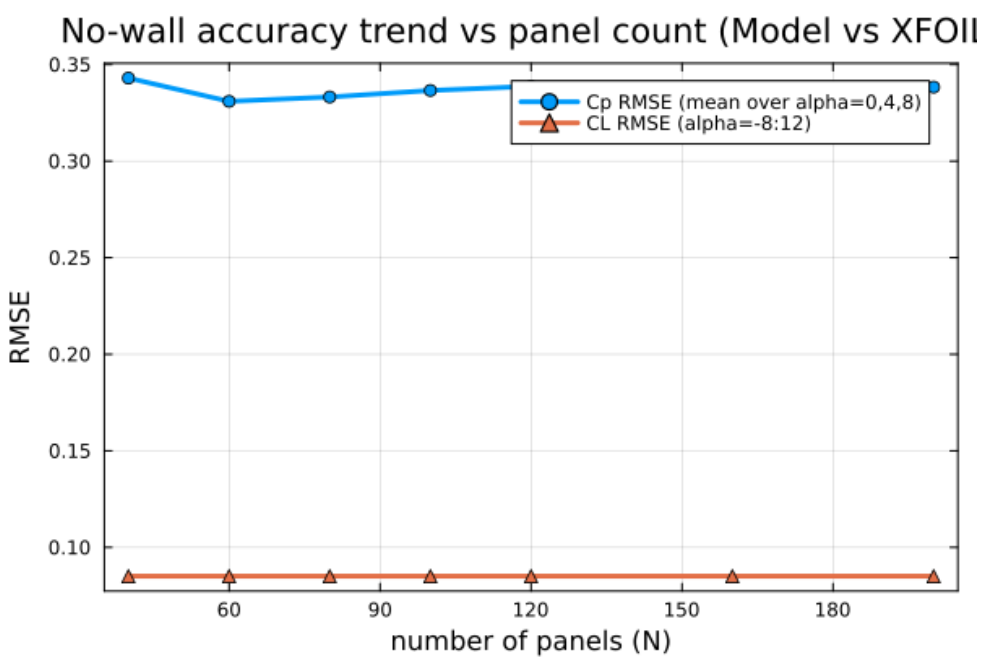

Figure 13: Legacy panel-count trend (camber-line model).

6 Discussion and Inferences

1. C_L increases with α for all heights.

2. The strongest ground effect appears as we move the airfoil closer to the ground (as intuitively

expected).

3. The updated surface-panel model significantly reduces the XFOIL mismatch compared with the older camber-line model.

4. A small residual error remains because the two solvers still differ in panel formulation, Kutta/wake treatment, and numerical details.

7 Model Limits

- This is a 2D inviscid potential-flow framework.
- No direct modeling of separation/stall physics.
- Ground is modeled as a slip wall using image panels.

8 Conclusion

After upgrading from camber-line to surface-panel modeling, the verification error against XFOIL is much smaller, which confirms the new equations are a better representation. At the same time, a small non-zero difference is still physically and numerically reasonable because the implementations are not identical.

Reference

M. Drela, "XFOIL: An Analysis and Design System for Low Reynolds Number Airfoils," in *Low Reynolds Number Aerodynamics*, Springer, Lecture Notes in Engineering 54, 1989.